

Medical Assistant Build Ledger

Bundles I through the final performance and calibration layers

This document packages the later-stage bundles of the platform into one PDF so it can be uploaded into Replit as a compact implementation reference. It focuses on what was added, why it matters, and which files belong to each bundle.

- Path 1: deeper clinical intelligence, scoring, Bayesian evidence, adaptive learning, and calibration.
- Path 2: organizations, notifications, audit/compliance, export packets, messaging orchestration, and deployment resilience.
- Governance: formulary, prescribing control, consent, validation, release governance, medication visibility protection.
- Final platform optimizations: architecture risk controls, orchestrator simplification, hot-path performance, and diagnostic calibration.

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Bundle I — Export Safety Layer

Adds pre-export safety checks for signoff, note draft, disposition, unresolved critical questions, red flags, and prior export state.

Files

server/services/caseExportReadinessChecker.ts
server/routes/exportReadiness.ts
client/src/components/ExportReadinessPanel.tsx

Bundle J — Physician Override Intelligence

Surfaces override patterns by complaint so the engine can be tuned where physicians repeatedly disagree with the model.

Files

server/services/overridePatternAnalyzer.ts
server/routes/overridePatterns.ts
client/src/pages/OverridePatterns.tsx

Bundle K — Question Gap Analyzer

Shows which clinically important questions are commonly missing at intake or later requested during review.

Files

server/services/questionGapAnalyzer.ts
server/routes/questionGaps.ts
client/src/pages/QuestionGaps.tsx

Bundle L — Ops Daily Digest

Creates a daily operational summary across cases, discrepancies, queue aging, and blocked exports.

Files

server/services/opsDailyDigestBuilder.ts
server/routes/opsDailyDigest.ts
client/src/pages/OpsDailyDigest.tsx

Bundle M — Clinical Workflow Health

Produces a top-level workflow health score from discrepancies, weak signoffs, and export blockers.

Files

server/services/clinicalWorkflowHealthService.ts
server/routes/clinicalWorkflowHealth.ts
client/src/pages/ClinicalWorkflowHealth.tsx

Bundle N — Case Ops Actions

Adds operational actions such as assign reviewer, request more info, escalate, and close case.

Files

server/services/caseOpsActionService.ts
server/routes/caseOpsActions.ts
client/src/components/CaseOpsActions.tsx

Bundle O — Real Engine Adapter Hardening

Unifies chat/case state mapping into the deterministic engine and normalizes traces for debugging and review.

Files

server/services/chatEngineStateManager.ts
server/services/chatEngineAdapterV2.ts
server/services/chatEngineTraceNormalizer.ts
server/routes/chatEngineDebug.ts

Implementation notes

- Hot-path or inference relevance: affects case-state mapping, question selection, probability updating, calibration, or latency.

Bundle P — Dynamic Question Selection

Ranks candidate questions by how much they separate the leading diagnoses, reducing wasted questioning.

Files

server/services/questionDiscriminatorScorer.ts
server/services/differentialSeparationEngine.ts
server/services/chatNextBestQuestionSelector.ts
server/routes/nextBestQuestionDebug.ts

Implementation notes

- Hot-path or inference relevance: affects case-state mapping, question selection, probability updating, calibration, or latency.

Bundle Q — Outcome Feedback Capture

Captures post-review outcomes, final diagnoses, medication/test changes, and outcome notes for later learning.

Files

server/types/outcome.ts
server/services/outcomeCaptureService.ts
server/routes/outcomes.ts

```
client/src/pages/Outcomes.tsx
```

Bundle R — Complaint QA Dashboards

Provides complaint-specific QA views for overrides, question gaps, export failures, and disposition distribution.

Files

```
server/services/complaintQADashboardService.ts
server/routes/complaintQA.ts
client/src/components/ComplaintQASummaryCard.tsx
client/src/pages/ComplaintQA.tsx
```

Bundle S — Scoring Systems as First-Class Modules

Adds structured, deterministic clinical scores that can be called by the engine and previewed in UI.

Files

```
server/services/scoring/centorScore.ts
server/services/scoring/wellsScore.ts
server/services/scoring/heartScore.ts
server/services/scoring/scoringRegistry.ts
server/routes/scoringPreview.ts
client/src/components/ScorePreviewPanel.tsx
```

Bundle T — Medication Safety + Clinical Constraint Engine

Imposes patient factors, drug interactions, dosing adjustments, objective data, and contextual constraints before medication preview.

Files

```
server/services/clinicalConstraints/patientConstraintEngine.ts
server/services/clinicalConstraints/drugInteractionChecker.ts
server/services/clinicalConstraints/renalDoseAdjuster.ts
server/services/clinicalConstraints/hepaticDoseAdjuster.ts
server/services/clinicalConstraints/qtRiskChecker.ts
server/services/clinicalConstraints/pregnancyMedicationChecker.ts
server/services/clinicalConstraints/objectiveDataInterpreter.ts
server/services/clinicalConstraints/contextModifierEngine.ts
server/services/clinicalConstraints/medicationRecommendationBuilder.ts
server/routes/medicationPreview.ts
client/src/components/MedicationPreviewPanel.tsx
```

Implementation notes

- Medication safety relevance: constrains visibility, formulary status, prescribing approval, or risk checking.

Bundle U — Real Organization + Production Auth

Introduces organization-aware deployment and production auth scaffolding for multi-clinic operation.

Files

server/services/firebaseAuthAdapter.ts
server/services/organizationService.ts
server/middleware/requireOrganization.ts
server/routes/organizations.ts
client/src/pages/OrganizationSettings.tsx

Implementation notes

- Production relevance: strengthens deployment, messaging, export packets, organization scoping, or resilience.

Bundle V — Notification and Escalation Workflow

Adds operational alerts for case assignment, escalation, queue aging, and workflow events.

Files

server/services/notificationService.ts
server/services/escalationNotificationRouter.ts
server/routes/notifications.ts
client/src/pages/Notifications.tsx

Bundle W — eCW Deeper Sidecar Workflow

Creates structured encounter packets and export manifests for traceable EHR-sidecar workflows.

Files

server/services/ecwEncounterPacketBuilder.ts
server/services/ecwExportManifestService.ts
server/routes/ecwPackets.ts
client/src/pages/EcwExportWorkbench.tsx

Implementation notes

- Production relevance: strengthens deployment, messaging, export packets, organization scoping, or resilience.

Bundle X — Audit & Compliance Tooling

Captures access and export events and generates admin-facing audit reports.

Files

server/services/accessLogService.ts
server/services/auditReportService.ts

server/routes/auditReports.ts
client/src/pages/AuditReports.tsx

Implementation notes

- Governance relevance: supports auditability, integrity, consent, release control, or architecture risk mitigation.

Bundle Y — Messaging & Channel Orchestration

Unifies inbound/outbound messaging across web, Telegram, internal staff views, and future channels.

Files

server/services/channelOrchestrator.ts
server/services/messageRoutingService.ts
server/routes/messages.ts
client/src/pages/MessageOps.tsx

Implementation notes

- Production relevance: strengthens deployment, messaging, export packets, organization scoping, or resilience.

Bundle Z — Production Deployment & Resilience

Adds health checks, background jobs, deployment instructions, and incident response guidance.

Files

server/services/healthcheckService.ts
server/routes/health.ts
server/services/jobRunner.ts
docs/DEPLOYMENT_RUNBOOK.md
docs/INCIDENT_RESPONSE.md

Implementation notes

- Production relevance: strengthens deployment, messaging, export packets, organization scoping, or resilience.

Bundle AA — Medication Safety Governance

Introduces formulary governance, medication rule registries, and administrative medication controls.

Files

server/services/medicationGovernance/formularyService.ts
server/services/medicationGovernance/medicationRuleRegistry.ts
server/services/medicationGovernance/medicationSafetyService.ts
server/routes/formulary.ts
client/src/pages/FormularyAdmin.tsx

Implementation notes

- Medication safety relevance: constrains visibility, formulary status, prescribing approval, or risk checking.

Bundle AB — Outcome Monitoring System

Tracks bouncebacks, ED transfers, hospitalizations, and diagnosis revisions as safety signals.

Files

server/services/outcomeMonitoring/bouncebackAnalyzer.ts
server/services/outcomeMonitoring/outcomeMonitoringService.ts
server/routes/outcomeMonitoring.ts
client/src/pages/OutcomeMonitoring.tsx

Bundle AC — Physician Prescribing Controls

Requires physician medication approval, logs prescribing decisions, and handles restricted classes.

Files

server/services/prescribingControlService.ts
server/services/prescribingAuditLog.ts
server/routes/prescribingControls.ts
client/src/components/PrescribingApprovalPanel.tsx

Implementation notes

- Medication safety relevance: constrains visibility, formulary status, prescribing approval, or risk checking.

Bundle AD — Patient Identity & Consent Workflow

Captures identity, treatment/data/communication consent, and consent audit history.

Files

server/services/patientIdentityService.ts
server/services/patientConsentService.ts
server/routes/patientConsent.ts
client/src/pages/PatientConsentAdmin.tsx

Implementation notes

- Governance relevance: supports auditability, integrity, consent, release control, or architecture risk mitigation.

Bundle AE — Clinical Validation Dashboards

Surfaces discrepancies, overrides, under-triage signals, and validation metrics prior to deployment.

Files

server/services/validation/clinicalValidationService.ts
server/routes/clinicalValidation.ts

client/src/components/ValidationMetricCard.tsx
client/src/pages/ClinicalValidation.tsx

Bundle AF — Release Governance

Adds rule versioning, rollout percentages, enable/disable flags, and canary-style release control.

Files

server/services/releaseGovernanceService.ts
server/routes/releaseGovernance.ts
client/src/components/ReleaseControlPanel.tsx
client/src/pages/ReleaseGovernance.tsx

Implementation notes

- Governance relevance: supports auditability, integrity, consent, release control, or architecture risk mitigation.

Bundle AG — Open-Source Agent Framework Integration

Lets external/open-source agent frameworks use platform tools for QA, case fetches, and validation.

Files

server/services/agents/toolRegistry.ts
server/services/agents/clinicalAgentTools.ts
server/services/agents/openSourceAgentAdapter.ts
server/routes/agentTasks.ts
client/src/pages/AgentOps.tsx

Implementation notes

- Agent/AI relevance: adds orchestration, tool use, or language-model assistance around the deterministic engine.

Bundle AH — Microsoft Agent Framework Integration

Adds Microsoft-style orchestration, memory, and task agents for reasoning, review, and chart support.

Files

server/services/agents/msAgentMemory.ts
server/services/agents/msAgentOrchestrator.ts
server/services/agents/msClinicalReasoningAgent.ts
server/services/agents/msReviewAgent.ts
server/services/agents/msChartAgent.ts
server/routes/msAgentTasks.ts
client/src/pages/MicrosoftAgentOps.tsx

Implementation notes

- Agent/AI relevance: adds orchestration, tool use, or language-model assistance around the deterministic engine.

Bundle AI — ChatGPT Integration Layer

Connects ChatGPT safely for summarization and drafting while preserving deterministic clinical control.

Files

server/services/ai/chatgptClient.ts
server/services/ai/promptTemplates.ts
server/services/ai/aiReasoningService.ts
server/services/ai/aiSafetyGuardrails.ts
server/routes/aiTasks.ts
client/src/pages/AIAssistant.tsx

Implementation notes

- Agent/AI relevance: adds orchestration, tool use, or language-model assistance around the deterministic engine.

Bundle AIQ — Differential-Driven Question Selection

Selects the most informative next question based on diagnosis separation and uncertainty reduction.

Files

server/services/diagnostic/differentialProbabilityEngine.ts
server/services/diagnostic/questionImpactAnalyzer.ts
server/services/diagnostic/nextBestQuestionEngine.ts
server/routes/questionImpactDebug.ts
client/src/pages/NextBestQuestionInspector.tsx

Implementation notes

- Hot-path or inference relevance: affects case-state mapping, question selection, probability updating, calibration, or latency.

Bundle BAE — Bayesian Evidence Engine

Updates diagnosis probabilities in real time from answers, labs, vitals, and test evidence.

Files

server/services/diagnostic/bayesianEvidenceEngine.ts
server/services/diagnostic/evidenceLikelihoodRegistry.ts
server/services/diagnostic/bayesianUpdater.ts
server/services/diagnostic/labEvidenceAdapter.ts
server/routes/bayesianDebug.ts
client/src/pages/BayesianInspector.tsx

Implementation notes

- Hot-path or inference relevance: affects case-state mapping, question selection, probability updating, calibration, or latency.

Bundle AEL — Adaptive Evidence Learning Engine

Learns evidence likelihood weights from accumulated outcomes and retrain them via background jobs.

Files

server/services/diagnostic/adaptiveEvidenceLearner.ts
server/services/diagnostic/evidenceLikelihoodStore.ts
server/services/diagnostic/evidenceLearningJob.ts
server/routes/evidenceLearning.ts
client/src/pages/EvidenceLearningDashboard.tsx

Implementation notes

- Hot-path or inference relevance: affects case-state mapping, question selection, probability updating, calibration, or latency.

Bundle CAL — Diagnostic Calibration Engine

Calibrates diagnosis probabilities against historical outcomes to correct systematic bias.

Files

server/services/diagnostic/calibrationStore.ts
server/services/diagnostic/calibrationTrainer.ts
server/services/diagnostic/calibrationApplier.ts
server/services/diagnostic/calibrationJob.ts
server/routes/calibration.ts
client/src/pages/CalibrationDashboard.tsx

Implementation notes

- Hot-path or inference relevance: affects case-state mapping, question selection, probability updating, calibration, or latency.

Bundle AR — Architecture Risk Mitigation

Mitigates configuration drift, latency spikes, and concurrent case update corruption.

Files

server/services/riskControls/configIntegrityService.ts
server/services/riskControls/engineLatencyGuard.ts
server/services/riskControls/caseConcurrencyGuard.ts
server/routes/systemIntegrity.ts
client/src/pages/SystemIntegrity.tsx

Implementation notes

- Governance relevance: supports auditability, integrity, consent, release control, or architecture risk mitigation.

Bundle MR — Medication Visibility Guard

Ensures medication-related guidance cannot be shown to patients and is restricted to physician review flows.

Files

server/services/riskControls/medicationVisibilityGuard.ts
server/middleware/enforceMedicationVisibility.ts
server/routes/medicationVisibilityAudit.ts
client/src/components/MedicationVisibilityNotice.tsx

Implementation notes

- Medication safety relevance: constrains visibility, formulary status, prescribing approval, or risk checking.

Bundle SO — System Orchestrator Simplification

Introduces a central orchestrator so interfaces call one workflow layer instead of many services directly.

Files

server/services/caseOrchestrator.ts

Bundle PERF — Performance Optimization Layer

Adds rule/config caches, expression compilation caches, case caching, and a hot-path executor for low latency.

Files

server/services/performance/ruleCacheService.ts
server/services/performance/expressionCacheService.ts
server/services/performance/caseCacheService.ts
server/services/performance/hotPathExecutor.ts
server/routes/performanceStats.ts

Implementation notes

- Hot-path or inference relevance: affects case-state mapping, question selection, probability updating, calibration, or latency.

Bundle CAL+ — Diagnostic Calibration Accuracy Upgrade

Final probability calibration layer that improves diagnostic accuracy without increasing hot-path latency.

Files

server/services/diagnostic/calibrationStore.ts
server/services/diagnostic/calibrationTrainer.ts
server/services/diagnostic/calibrationApplier.ts
server/services/diagnostic/calibrationJob.ts

server/routes/calibration.ts
client/src/pages/CalibrationDashboard.tsx

Implementation notes

- Hot-path or inference relevance: affects case-state mapping, question selection, probability updating, calibration, or latency.

High-volume testing recommendation

Yes: the platform can and should be tested much more aggressively than a passive six-month observation window. The fastest route is a blended validation stack that combines massive synthetic regression with staged real-world shadow-mode comparison against your own decisions.

Recommended rapid-validation stack

- Mass synthetic case generation for each complaint, including modifiers, red flags, contradictory findings, and missing-data scenarios.
- Clinician-authored gold sets for high-risk complaints to anchor expected diagnosis, disposition, and question sequence.
- Monte Carlo perturbation of vitals, labs, and durations to detect unstable boundaries and brittle rules.
- Automated mismatch triage dashboards: under-triage, over-triage, medication safety block misses, and question-efficiency misses.
- Shadow-mode replay against your actual decision patterns, then complaint-by-complaint refinement.

What synthetic volume can do

Running millions of cases is realistic for deterministic bundles if the generator and harness are built efficiently. That will accelerate bug discovery, edge-case coverage, and calibration, especially for question selection, Bayesian evidence, and release governance. The caveat is that synthetic scale does not replace real workflow validation, because patient behavior, messy documentation, sparse data, and clinical nuance still matter in production.